

GATE DA (Data Science & AI) Practice Question Bank - Set 2

7 Topics · Previous-Year-Style + New Practice Questions · Full Solutions

This set covers seven topics: Discrete Random Variables & PMF, Multiple Linear Regression (Gradient Descent & Normal Equation), Relational Model Keys & Constraints, Averages, Recursion Depth & Call-Stack Tracing, Recursion Patterns (Factorial / Fibonacci / Recurrence-writing / GCD), and Uninformed Search -- BFS (mechanism & properties).

SAME HONESTY POLICY AS BEFORE: GATE DA has run only in 2024, 2025 and 2026, and it is a non-coding, MCQ/MSQ/NAT written exam. Every "old" question below is tagged with its real provenance:

[VERIFIED PYQ] -- a genuine, well-documented past GATE question (year and paper stated). Where the source is GATE CSE rather than GATE DA specifically, that is stated explicitly -- these test the identical concept/format used in GATE DA's shared syllabus (Probability & Statistics, Linear Algebra/ML, DBMS, Programming & Data Structures, and AI/Search are common to both).

[GATE-style] -- not a verbatim past question (either no verified PYQ exists on that exact sub-point, or the original figure/data could not be reliably reproduced), but written to match real GATE DA difficulty, phrasing, and mark-weightage.

Section B ("New Practice Questions") in every topic is freshly written GATE-level practice, as requested -- none of these are claimed to be past questions.

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Topic 1: Discrete Random Variable & PMF

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CSE 2024, Set 2 (same computation, reframed with an explicit random variable)]

Q1. [MCQ · 2 marks]

Six unbiased dice are rolled simultaneously. Let X denote the number of distinct face values that appear among the six outcomes. What is $P(X = 6)$, i.e. the probability that all six numbers 1 through 6 appear exactly once?

- (A) $5/324$
- (B) $1/324$
- (C) $5/1296$
- (D) $1/6$

Answer: (A)

Solution: Total equally-likely outcomes = $6^6 = 46656$. Favourable outcomes (all six faces distinct) = $6! = 720$ (each die gets a different face -- a permutation of $\{1, \dots, 6\}$). $P(X=6) = 720/46656 = 5/324$.

[GATE-style, classic and widely used in GATE-level probability practice]

Q2. [NAT · 2 marks]

A discrete random variable X has PMF $P(X=x) = 1/2^x$ for $x = 1, 2, 3, \dots$. Find $P(X > 4)$.

Answer: 0.0625

Solution: $P(X > 4) = 1 - P(X \leq 4) = 1 - (1/2 + 1/4 + 1/8 + 1/16) = 1 - 15/16 = 1/16 = 0.0625$.

[GATE-style]

Q3. [NAT · 1 mark]

A discrete random variable X takes values $\{0, 1, 2, 3\}$ with $P(X=0)=0.2$, $P(X=1)=0.3$, $P(X=2)=k$, $P(X=3)=0.2$. Find k .

Answer: 0.30

Solution: Since probabilities must sum to 1: $0.2 + 0.3 + k + 0.2 = 1 \Rightarrow k = 0.30$.

[GATE-style]

Q4. [NAT · 2 marks]

Using the PMF from the previous question (with $k=0.3$), find $E[X]$.

Answer: 1.50

Solution: $E[X] = 0(0.2) + 1(0.3) + 2(0.3) + 3(0.2) = 0 + 0.3 + 0.6 + 0.6 = 1.50$.

[GATE-style]

Q5. [NAT · 2 marks]

Using the same PMF ($P(X=0)=0.2$, $P(X=1)=0.3$, $P(X=2)=0.3$, $P(X=3)=0.2$), find $\text{Var}(X)$.

Answer: 1.05

Solution: $E[X^2] = 0(0.2) + 1(0.3) + 4(0.3) + 9(0.2) = 0 + 0.3 + 1.2 + 1.8 = 3.3$.
 $\text{Var}(X) = E[X^2] - (E[X])^2 = 3.3 - (1.5)^2 = 3.3 - 2.25 = 1.05$.

[GATE-style]

Q6. [MCQ · 1 mark]

For a fair six-sided die, let X be the outcome of a single roll. What is $E[X]$?

- (A) 3
- (B) 3.5
- (C) 4
- (D) 21

Answer: (B)

Solution: $E[X] = (1+2+3+4+5+6)/6 = 21/6 = 3.5$.

[GATE-style]

Q7. [MCQ · 1 mark]

Which of the following is a valid PMF for a discrete random variable taking values in $\{1,2,3\}$?

- (A) 0.2, 0.5, 0.3
- (B) 0.5, 0.5, 0.5
- (C) -0.1 , 0.6, 0.5
- (D) 0.3, 0.3, 0.3

Answer: (A)

Solution: A valid PMF requires all probabilities ≥ 0 and summing to 1. Only (A): $0.2+0.5+0.3=1$, all non-negative.

[GATE-style]

Q8. [MSQ · 2 marks]

Which of the following properties must every valid PMF $p(x)$ of a discrete random variable satisfy?

- (A) $p(x) \geq 0$ for every x
- (B) $\sum p(x) = 1$, summed over the support
- (C) $p(x)$ is non-decreasing as x increases
- (D) $p(x) \leq 1$ for every individual x

Answer: (A), (B), (D)

Solution: (C) is false -- there is no requirement that the PMF be monotonic in x ; it can rise and fall arbitrarily.

[GATE-style]

Q9. [NAT · 1 mark]

Two fair coins are tossed independently. Let X be the number of heads obtained. Find $P(X=1)$.

Answer: 0.50

Solution: Sample space $\{HH, HT, TH, TT\}$, each with probability $1/4$. $X=1$ for outcomes HT and TH $\Rightarrow P(X=1)=2/4=0.5$.

[GATE-style]

Q10. [NAT · 2 marks]

Let $X \sim \text{Binomial}(n=4, p=0.5)$. Find $P(X=2)$.

Answer: 0.375

Solution: $P(X=2) = C(4,2)(0.5)^2(0.5)^2 = 6 \times 0.25 \times 0.25 = 6 \times 0.0625 = 0.375$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 2 marks]

A discrete RV X has PMF $P(X=1)=0.1$, $P(X=2)=0.4$, $P(X=3)=0.5$. Find $E[X]$.

Answer: 2.40

Solution: $E[X] = 1(0.1)+2(0.4)+3(0.5) = 0.1+0.8+1.5 = 2.40$.

[New Practice]

Q12. [NAT · 1 mark]

Using the PMF above, find $E[X^2]$.

Answer: 6.20

Solution: $E[X^2] = 1(0.1)+4(0.4)+9(0.5) = 0.1+1.6+4.5 = 6.20$.

[New Practice]

Q13. [NAT · 1 mark]

Using the results of the previous two questions, find $\text{Var}(X)$.

Answer: 0.44

Solution: $\text{Var}(X) = E[X^2] - (E[X])^2 = 6.2 - (2.4)^2 = 6.2 - 5.76 = 0.44$.

[New Practice]

Q14. [MCQ · 1 mark]

For a discrete random variable uniformly distributed over $\{1,2,\dots,n\}$, $E[X]$ equals:

- (A) $n/2$
- (B) $(n+1)/2$
- (C) n
- (D) $(n-1)/2$

Answer: (B)

Solution: For discrete uniform on $\{1,\dots,n\}$, $E[X] = (1+2+\dots+n)/n = [n(n+1)/2]/n = (n+1)/2$.

[New Practice]

Q15. [NAT · 1 mark]

Using the formula from the previous question, find $E[X]$ for a discrete random variable uniform on $\{1,\dots,5\}$.

Answer: 3

Solution: $(n+1)/2 = (5+1)/2 = 3$.

[New Practice]

Q16. [MCQ · 1 mark]

If $X \sim \text{Bernoulli}(p)$, then $\text{Var}(X)$ equals:

- (A) p
- (B) $p(1-p)$
- (C) $1-p$
- (D) p^2

Answer: (B)

Solution: For Bernoulli(p): $E[X]=p$, $E[X^2]=p$, so $\text{Var}(X)=p-p^2=p(1-p)$.

[New Practice]

Q17. [NAT · 1 mark]

A biased coin has $P(\text{Head}) = 0.3$. Let $X=1$ if heads, $X=0$ if tails. Find $\text{Var}(X)$.

Answer: 0.21

Solution: $\text{Var}(X) = p(1-p) = 0.3 \times 0.7 = 0.21$.

[New Practice]

Q18. [MSQ · 2 marks]

Which of the following are TRUE about the cumulative distribution function $F(x)=P(X \leq x)$ of a discrete random variable?

- (A) F is non-decreasing
- (B) F is right-continuous, with jumps at the values in the support
- (C) $\lim_{x \rightarrow -\infty} F(x)$ equals 1
- (D) $\lim_{x \rightarrow +\infty} F(x)$ equals 1

Answer: (A), (B), (D)

Solution: (C) is false -- $F(x)$ must approach 0, not 1, as $x \rightarrow -\infty$.

[New Practice]

Q19. [NAT · 1 mark]

A discrete RV X takes values $\{-1, 0, 1\}$, each with equal probability $1/3$. Find $E[X]$.

Answer: 0

Solution: $E[X] = (-1+0+1)/3 = 0/3 = 0$.

[New Practice]

Q20. [NAT · 2 marks]

For the random variable in the previous question, find $\text{Var}(X)$ (correct to 2 decimals).

Answer: 0.67

Solution: $E[X^2] = (1+0+1)/3 = 2/3$. $\text{Var}(X) = E[X^2] - (E[X])^2 = 2/3 - 0 = 0.67$ (accept 0.66-0.67).

Topic 2: Multiple Linear Regression - Gradient Descent & Normal Equation

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

For multiple linear regression $y = Xw$ (X is the $m \times n$ design matrix, including an intercept column; w is the $n \times 1$ parameter vector), the least-squares normal equation solution is:

- (A) $w = (X^T X)^{-1} X^T y$
- (B) $w = (X X^T)^{-1} X^T y$
- (C) $w = X^T (X X^T)^{-1} y$
- (D) $w = X^{-1} y$

Answer: (A)

Solution: Setting the gradient of $J(w) = (1/2m) \|Xw - y\|^2$ to zero gives $X^T X w = X^T y$, so $w = (X^T X)^{-1} X^T y$, provided $X^T X$ is invertible.

[GATE-style]

Q2. [MCQ · 1 mark]

The normal equation $w = (X^T X)^{-1} X^T y$ requires $X^T X$ to be:

- (A) Invertible (non-singular)
- (B) Diagonal
- (C) Symmetric only
- (D) Sparse

Answer: (A)

Solution: $X^T X$ is always symmetric, but computing its inverse additionally requires it to be non-singular, i.e. X must have full column rank.

[GATE-style]

Q3. [MCQ · 2 marks]

For the cost function $J(w) = (1/2m) \|Xw - y\|^2$, the gradient descent update rule for multiple linear regression is:

- (A) $w := w - \alpha(1/m) X^T (Xw - y)$
- (B) $w := w - \alpha(1/m) X (Xw - y)$
- (C) $w := w + \alpha(1/m) X^T (Xw - y)$
- (D) $w := w - \alpha X^T X w$

Answer: (A)

Solution: $\nabla J(w) = (1/m) X^T (Xw - y)$. Gradient descent moves against the gradient: $w := w - \alpha \nabla J(w)$.

[GATE-style]

Q4. [NAT · 1 mark]

Given $X^T X = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$, compute $\det(X^T X)$ to check invertibility.

Answer: 8

Solution: $\det = (4)(3) - (2)(2) = 12 - 4 = 8$.

[GATE-style]

Q5. [MCQ · 1 mark]

If the number of features n exceeds the number of training examples m ($n > m$), the $n \times n$ matrix $X^T X$ is:

- (A) Invertible
- (B) Singular (not invertible)
- (C) Diagonal
- (D) Always the identity matrix

Answer: (B)

Solution: $\text{rank}(X^T X) = \text{rank}(X) \leq \min(m, n) = m < n$, so an $n \times n$ matrix with $\text{rank} < n$ must be singular.

[GATE-style]

Q6. [MSQ · 2 marks]

Which of the following are TRUE when comparing gradient descent and the normal equation for multiple linear regression?

- (A) The normal equation gives an exact closed-form solution in a single computation (no iterations)
- (B) Gradient descent requires choosing a learning rate α
- (C) The normal equation remains computationally cheap even when the number of features n is very large
- (D) Gradient descent can still be applied even when $X^T X$ is singular or ill-conditioned

Answer: (A), (B), (D)

Solution: (C) is false -- computing $(X^T X)^{-1}$ costs $O(n^3)$, which becomes expensive as n grows large; gradient descent scales much better in that regime.

[GATE-style]

Q7. [NAT · 2 marks]

For $X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ (intercept column + one feature), compute $\det(X^T X)$.

Answer: 6

Solution: $X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$ (column sums: $\sum 1=3$, $\sum x=6$, $\sum x^2=14$). $\det = 3 \times 14 - 6 \times 6 = 42 - 36 = 6$.

[GATE-style]

Q8. [MCQ · 1 mark]

In gradient descent for multiple linear regression, feature scaling (normalization) is mainly done to:

- (A) Reduce the number of features
- (B) Speed up convergence by making the cost function's contours more circular (less elongated)
- (C) Guarantee a global minimum that would not otherwise exist
- (D) Eliminate the need to choose a learning rate

Answer: (B)

[GATE-style]

Q9. [MCQ · 1 mark]

Which of the following is most likely to cause gradient descent for linear regression to diverge (cost increasing instead of decreasing)?

- (A) Learning rate too small
- (B) Learning rate too large
- (C) Too many training examples
- (D) Using the normal equation instead

Answer: (B)

[GATE-style]

Q10. [NAT · 1 mark]

Computing $(X^T X)^{-1}$ for an $n \times n$ matrix via standard matrix inversion has time complexity $O(n^k)$ for the dominant term. What is k ?

Answer: 3

Solution: Standard (Gauss-Jordan / LU-based) matrix inversion of an $n \times n$ matrix costs $O(n^3)$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [MCQ · 1 mark]

In the design matrix X for multiple linear regression, the column of all 1's represents:

- (A) The feature with the highest variance
- (B) The intercept (bias) term
- (C) The target variable
- (D) A regularization term

Answer: (B)

[New Practice]

Q12. [NAT · 1 mark]

For $X = [[1,2],[1,3],[1,4],[1,5]]$ and $y = [3,5,6,9]$, find the first component of $X^T y$ (i.e. $\sum y_i$).

Answer: 23

Solution: $\sum y_i = 3+5+6+9 = 23$.

[New Practice]

Q13. [NAT · 2 marks]

Using the same data, find the second component of $X^T y$ (i.e. $\sum x_i y_i$, where x is the second column: 2,3,4,5).

Answer: 90

Solution: $\sum x_i y_i = 2(3)+3(5)+4(6)+5(9) = 6+15+24+45 = 90$.

[New Practice]

Q14. [MCQ · 1 mark]

With a suitably chosen (not too large) learning rate, the cost $J(w)$ over successive gradient descent iterations should:

- (A) Increase monotonically
- (B) Decrease monotonically (or non-increasingly) toward a minimum
- (C) Oscillate indefinitely
- (D) Remain exactly constant

Answer: (B)

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following typically help gradient descent converge faster or more reliably for multiple linear regression?

- (A) Feature scaling / normalization
- (B) Choosing a well-tuned learning rate
- (C) Increasing the learning rate without any bound
- (D) Mean-centering the features

Answer: (A), (B), (D)

[New Practice]

Q16. [NAT · 1 mark]

Given $X^T X = \begin{bmatrix} 9 & 3 \\ 3 & 5 \end{bmatrix}$, find $\det(X^T X)$.

Answer: 36

Solution: $\det = (9)(5) - (3)(3) = 45 - 9 = 36$.

[New Practice]

Q17. [MCQ · 1 mark]

When $X^T X$ is invertible, the normal-equation solution $w = (X^T X)^{-1} X^T y$ is guaranteed to be:

- (A) The unique global minimizer of the least-squares cost function
- (B) Only a local minimum
- (C) Not necessarily better than a random guess
- (D) Valid only when $m < n$

Answer: (A)

Solution: Since $J(w)$ is convex, the unique stationary point (where the gradient is zero) is guaranteed to be the global minimum.

[New Practice]

Q18. [MCQ · 1 mark]

Batch gradient descent, as opposed to stochastic gradient descent, computes the gradient using:

- (A) A single randomly chosen training example per update
- (B) The entire training set per update
- (C) No training data at all
- (D) Only the first and last training examples

Answer: (B)

[New Practice]

Q19. [NAT · 1 mark]

A multiple linear regression model has 9 features (plus an intercept term). How many total parameters (entries of w) does it have?

Answer: 10

Solution: 9 features + 1 intercept = 10 parameters.

[New Practice]

Q20. [MCQ · 2 marks]

For a dataset with a very large number of features (e.g. $n = 100,000$), gradient descent is generally preferred over the normal equation mainly because:

- (A) Gradient descent avoids computing/inverting the $n \times n$ matrix $X^T X$, which becomes very expensive for large n
- (B) Gradient descent always gives an exact solution in a single step
- (C) Gradient descent needs no hyperparameters at all
- (D) Gradient descent only works when $n > m$

Answer: (A)

Topic 3: Relational Model – Keys & Constraints

Section A — Old / Previous-Year-Style Questions (10)

[VERIFIED PYQ — GATE CSE 2024, Set 1, Question 12]

Q1. [MSQ · 2 marks]

Consider a relation $R(\text{Name, Age, Address})$ where the composite attribute set $\{\text{Name, Age}\}$ is the primary key. Which of the following statements is/are TRUE?

- (A) R can be in 1NF despite having a composite (multi-attribute) primary key
- (B) R cannot have a foreign key
- (C) R cannot have a composite (multi-valued) attribute if R is in 1NF
- (D) R can have at most one candidate key

Answer: (A), (C)

Solution: (A) is true -- 1NF only requires atomic attribute values; a multi-attribute (composite) primary key is perfectly allowed. (B) is false -- 1NF places no restriction on foreign keys. (C) is true -- 1NF explicitly disallows composite/multi-valued attributes; every attribute value must be atomic. (D) is false -- a relation can have several candidate keys; one of them is simply chosen as the primary key.

[VERIFIED PYQ — GATE CS, classic referential-integrity question]

Q2. [MCQ · 2 marks]

Let $R(a,b,c)$ and $S(d,e,f)$ be two relations in which d is a foreign key of S that refers to the primary key of R . Consider the following operations: I. Insert into R , II. Insert into S , III. Delete from R , IV. Delete from S . Which of these operations can potentially cause a violation of the referential integrity constraint above?

- (A) I and II
- (B) II and III
- (C) I and III
- (D) II and IV

Answer: (B)

Solution: Inserting into R (I) only adds new primary-key values -- it cannot break any existing reference. Deleting from S (IV) only removes references -- it cannot create a dangling reference either. But inserting into S (II) with a foreign-key value not present in R violates referential integrity, and deleting a row from R (III) that is still referenced by some row in S also violates it (unless a cascade/set-null action is defined). So the answer is II and III.

[GATE-style, standard cascade-delete mechanics]

Q3. [NAT · 1 mark]

Table T_1 has attribute Q as a foreign key referencing attribute R of table T_2 , with ON DELETE CASCADE. There are exactly 3 rows in T_1 with $Q = 8$. If the row with $T_2.R = 8$ is deleted, how many additional rows are deleted from T_1 as a consequence of the cascade?

Answer: 3

Solution: ON DELETE CASCADE deletes every child row referencing the deleted parent key -- here, all 3 matching rows in T_1 .

[GATE-style]

Q4. [MCQ · 1 mark]

A candidate key of a relation is defined as:

- (A) Any attribute whose values happen to all be distinct
- (B) A minimal superkey (a superkey with no proper subset that is itself a superkey)
- (C) The primary key selected by the database administrator
- (D) Any set of attributes referencing another table

Answer: (B)

[GATE-style]

Q5. [MSQ · 2 marks]

Which of the following statements about keys in the relational model are TRUE?

- (A) Every relation must have at least one candidate key
- (B) A relation can have more than one candidate key
- (C) A primary key attribute's value can be NULL
- (D) A foreign key attribute's value can be NULL (if the foreign key column itself is not required to be non-null)

Answer: (A), (B), (D)

Solution: (C) is false -- the entity integrity constraint requires that no part of a primary key be NULL.

[GATE-style]

Q6. [MCQ · 1 mark]

The **entity integrity** constraint states that:

- (A) A non-null foreign key value must match an existing primary key value in the referenced relation
- (B) No attribute that is part of the primary key may be NULL
- (C) Every relation must have exactly one foreign key
- (D) All attributes of a relation must be atomic

Answer: (B)

[GATE-style]

Q7. [MCQ · 1 mark]

The **referential integrity** constraint states that:

- (A) A primary key value can never be NULL
- (B) A non-null foreign key value must match an existing primary key value in the referenced relation
- (C) Every attribute in a relation must be unique
- (D) A table cannot have more than one foreign key

Answer: (B)

[GATE-style]

Q8. [NAT · 1 mark]

A relation R(A,B,C,D) has exactly two candidate keys: {A,B} and {C,D} (both composite, minimal, and non-overlapping). What is the number of attributes in each candidate key?

Answer: 2

[GATE-style]

Q9. [MCQ · 1 mark]

A superkey that is NOT a candidate key is best described as:

- (A) A superkey that does not exist for any relation

- (B) A superkey that contains at least one candidate key as a proper subset, and hence is not minimal
- (C) A superkey that is always identical to the primary key
- (D) A superkey with exactly one attribute

Answer: (B)

[GATE-style]

Q10. [MCQ · 2 marks]

If {A,B} is a candidate key of relation R, which of the following statements is necessarily FALSE?

- (A) {A,B,C} is a superkey of R
- (B) {A} alone is also a superkey of R
- (C) {A,B} functionally determines every other attribute of R
- (D) R could have another, disjoint candidate key

Answer: (B)

Solution: If {A} alone were a superkey, then {A,B} would not be minimal, contradicting the fact that {A,B} is given to be a candidate key. So statement (B) must be false.

Section B — New Practice Questions (10)

[New Practice]

Q11. [MCQ · 1 mark]

In relation Employee(EmpID, Name, DeptID), if DeptID references Department(DeptID), then DeptID in Employee is called a:

- (A) Candidate key
- (B) Primary key
- (C) Foreign key
- (D) Super key

Answer: (C)

[New Practice]

Q12. [MSQ · 2 marks]

A relation has attributes A, B, C, with {A} being its only candidate key, and it is given that {B,C} alone does not determine the whole tuple. Which of the following are valid superkeys of this relation?

- (A) {A}
- (B) {A,B}
- (C) {A,B,C}
- (D) {B,C}

Answer: (A), (B), (C)

Solution: Any superset of a candidate key is a superkey; {B,C} is not, since it was stated not to determine the whole tuple.

[New Practice]

Q13. [NAT · 1 mark]

A relation has 4 attributes. How many distinct 2-attribute subsets does it have (i.e. $C(4,2)$), regardless of whether each forms a key?

Answer: 6

Solution: $C(4,2) = 4!/(2!2!) = 6$.

[New Practice]

Q14. [MCQ · 1 mark]

Given Orders.CustomerID references Customers.CustomerID, which of the following actions would violate referential integrity?

- (A) Inserting a new customer with a new CustomerID
- (B) Inserting an order with a CustomerID that does not exist in Customers
- (C) Deleting an order record
- (D) Updating a customer's name

Answer: (B)

[New Practice]

Q15. [MCQ · 1 mark]

The "ON DELETE CASCADE" referential action means that when a referenced (parent) row is deleted:

- (A) The delete operation is rejected
- (B) All matching child rows are automatically deleted as well
- (C) The foreign key value in matching child rows is set to NULL
- (D) Nothing happens to the child rows

Answer: (B)

[New Practice]

Q16. [MCQ · 1 mark]

The "ON DELETE SET NULL" referential action means that when a referenced (parent) row is deleted:

- (A) The delete operation is rejected
- (B) All matching child rows are deleted
- (C) The foreign key value in matching child rows is set to NULL
- (D) A new parent row is automatically created

Answer: (C)

[New Practice]

Q17. [MSQ · 2 marks]

Which of the following must hold for an attribute set K to be a candidate key of relation R?

- (A) K functionally determines all attributes of R (uniqueness)
- (B) No proper subset of K functionally determines all attributes of R (minimality)
- (C) K must consist of a single attribute
- (D) K must include the first attribute listed in R's schema

Answer: (A), (B)

[New Practice]

Q18. [NAT · 1 mark]

Relation R(A,B,C,D,E) has {A,B} as its only candidate key. A child table references R via a foreign key matching this candidate key. How many attributes does this foreign key have?

Answer: 2

[New Practice]

Q19. [MCQ · 2 marks]

Which of the following statements about NULLs and keys is TRUE?

- (A) A primary key attribute may contain NULL as long as its non-NULL values are unique
- (B) An alternate key (a candidate key not chosen as primary) can be enforced via a UNIQUE constraint, though plain SQL UNIQUE constraints typically still permit NULLs
- (C) A foreign key attribute can never be NULL under any circumstance
- (D) NULL values are guaranteed to be excluded from every uniqueness check, in every SQL implementation

Answer: (B)

Solution: (A) violates entity integrity (PK attributes must be entirely NOT NULL). (C) is false -- an optional foreign key can be NULL. (D) overgeneralizes NULL-handling behaviour, which is implementation-dependent.

[New Practice]

Q20. [MCQ · 1 mark]

Relation Student(RollNo, Email, Name, Dept) has two candidate keys: {RollNo} and {Email}. If {RollNo} is chosen as the primary key, {Email} is called a(n):

- (A) Foreign key
- (B) Super key only, not a candidate key
- (C) Alternate key
- (D) Composite key

Answer: (C)

Topic 4: Average

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

The average of 10 numbers is 15. If one of the numbers, 25, is removed, what is the new average of the remaining 9 numbers (correct to 2 decimals)?

- (A) 13.89
- (B) 15.00
- (C) 12.50
- (D) 14.00

Answer: (A)

Solution: Original sum = 150. Removing 25: new sum = 125. New average = $125/9 \approx 13.89$.

[GATE-style]

Q2. [NAT · 1 mark]

The average of 5 numbers is 20. A sixth number is added and the new average becomes 22. Find the sixth number.

Answer: 32

Solution: Sum of 5 numbers = 100. Sum of 6 numbers = $22 \times 6 = 132$. Sixth number = $132 - 100 = 32$.

[GATE-style]

Q3. [MCQ · 2 marks]

The average weight of 8 people increases by 2.5 kg when a new person replaces one of them who weighed 60 kg. Find the weight of the new person.

- (A) 80 kg
- (B) 70 kg
- (C) 65 kg
- (D) 75 kg

Answer: (A)

Solution: Total weight increases by $8 \times 2.5 = 20$ kg. So new person's weight = $60 + 20 = 80$ kg.

[GATE-style]

Q4. [NAT · 1 mark]

Find the average of the first 10 natural numbers.

Answer: 5.5

Solution: Sum = 55, count = 10, average = $55/10 = 5.5$.

[GATE-style]

Q5. [MCQ · 1 mark]

The average of three consecutive even numbers is 34. Find the largest of the three numbers.

- (A) 32
- (B) 34
- (C) 36
- (D) 38

Answer: (C)

Solution: Let the numbers be $x-2$, x , $x+2$. Their average is $x = 34$, so the largest is $x+2 = 36$.

[GATE-style]

Q6. [NAT · 2 marks]

The average marks of a class of 40 students is 60. The average of 15 of these students is 50. Find the average of the remaining 25 students.

Answer: 66

Solution: Total = $40 \times 60 = 2400$. Sum of 15 students = $15 \times 50 = 750$. Remaining sum = $2400 - 750 = 1650$. Average of remaining 25 = $1650/25 = 66$.

[GATE-style]

Q7. [MCQ · 1 mark]

A cricketer's average score after 10 innings is 40. After scoring 62 in the 11th innings, his new average becomes:

- (A) 40
- (B) 41
- (C) 42
- (D) 43

Answer: (C)

Solution: Total after 10 innings = 400. New total = $400 + 62 = 462$. New average = $462/11 = 42$.

[GATE-style]

Q8. [MSQ · 2 marks]

Which of the following statements about the arithmetic mean (average) are TRUE?

- (A) The mean is always one of the values in the dataset
- (B) The sum of deviations of all data points from the mean is always zero
- (C) The mean can be significantly affected by outliers
- (D) The mean of n numbers equals their sum divided by n

Answer: (B), (C), (D)

Solution: (A) is false -- e.g. the mean of $\{1,2\}$ is 1.5, which is not in the dataset.

[GATE-style]

Q9. [NAT · 1 mark]

The average of 7 consecutive integers is 20. Find the smallest of the seven integers.

Answer: 17

Solution: For an odd count of consecutive integers, the average equals the middle term, so the middle term is 20. The seven integers are 17,18,19,20,21,22,23 -- smallest is 17.

[GATE-style]

Q10. [MCQ · 1 mark]

If the average of x and y is 25, and the average of y and z is 35, what is $z-x$?

- (A) 10
- (B) 15
- (C) 20
- (D) 25

Answer: (C)

Solution: $x+y = 50$, $y+z = 70$. Subtracting: $(y+z)-(x+y) = z-x = 70-50 = 20$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

The average of 6 numbers is 18. If one number is excluded, the average of the remaining 5 becomes 20. Find the excluded number.

Answer: 8

Solution: Sum of 6 = 108. Sum of 5 = 100. Excluded number = $108-100 = 8$.

[New Practice]

Q12. [MCQ · 1 mark]

The average of the first n natural numbers is:

- (A) $n/2$
- (B) $(n+1)/2$
- (C) $n(n+1)/2$
- (D) $(n-1)/2$

Answer: (B)

[New Practice]

Q13. [NAT · 1 mark]

Using the formula from the previous question, find the average of the first 20 natural numbers.

Answer: 10.5

Solution: $(n+1)/2 = 21/2 = 10.5$.

[New Practice]

Q14. [MCQ · 2 marks]

The average age of a family of 5 members is 24 years. Excluding the youngest member, the average age of the remaining 4 becomes 27. Find the age of the youngest member.

- (A) 10
- (B) 12
- (C) 14
- (D) 16

Answer: (B)

Solution: Sum of 5 = 120. Sum of 4 = 108. Youngest member's age = $120-108 = 12$.

[New Practice]

Q15. [MSQ · 2 marks]

Which of the following statements about how transformations affect the mean of a dataset are TRUE?

- (A) Adding a constant c to every value shifts the mean by c
- (B) Multiplying every value by a constant k scales the mean by k
- (C) Squaring every value leaves the mean numerically unchanged

(D) Replacing every value with the original mean results in a new dataset whose mean equals the original mean

Answer: (A), (B), (D)

Solution: (C) is false in general -- the mean of the squares is not equal to the square of the mean (except in special cases).

[New Practice]

Q16. [NAT · 1 mark]

Find the mean of the dataset: 4, 8, 6, 10, 12.

Answer: 8

Solution: Sum = 40, $n = 5$, mean = $40/5 = 8$.

[New Practice]

Q17. [MCQ · 2 marks]

A class has two groups: Group 1 has 20 students with average score 60; Group 2 has 30 students with average score 80. Find the combined average score of all 50 students.

- (A) 70
- (B) 72
- (C) 74
- (D) 75

Answer: (B)

Solution: Combined average = $(20 \times 60 + 30 \times 80)/50 = (1200 + 2400)/50 = 3600/50 = 72$.

[New Practice]

Q18. [NAT · 1 mark]

The average of 4 numbers is 50. Three of the numbers are 40, 55, and 60. Find the fourth number.

Answer: 45

Solution: Sum of 4 = 200. Sum of first three = 155. Fourth number = $200 - 155 = 45$.

[New Practice]

Q19. [MCQ · 2 marks]

For a round trip of equal distance each way, the outbound and return speeds are 40 km/h and 60 km/h respectively. The average speed for the entire round trip is:

- (A) 48 km/h
- (B) 50 km/h
- (C) 52 km/h
- (D) 45 km/h

Answer: (A)

Solution: For equal distances, the average speed is the harmonic mean: $2 \times 40 \times 60 / (40 + 60) = 4800/100 = 48$ km/h (NOT the simple arithmetic mean of 50).

[New Practice]

Q20. [NAT · 2 marks]

The average of 9 numbers is 35. The average of the first 5 of them is 30. Find the average of the remaining 4 numbers (correct to 2 decimals).

Answer: 41.25

Solution: Total sum = $9 \times 35 = 315$. Sum of first 5 = $5 \times 30 = 150$. Sum of remaining 4 = $315 - 150 = 165$. Average = $165/4 = 41.25$.

Topic 5: Recursion Depth & Call-Stack Tracing

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style, activation-record/recursion-depth concept classically tested in GATE]

Q1. [NAT · 1 mark]

Consider `fact(n){ if(n<=1) return 1; return n*fact(n-1); }`. What is the maximum recursion depth (number of activation records simultaneously on the stack, including the initial call) when computing `fact(5)`?

Answer: 5

Solution: The call chain is `fact(5)→fact(4)→fact(3)→fact(2)→fact(1)`, which stops at the base case `fact(1)`. That is 5 nested activation records (for `n=5,4,3,2,1`).

[GATE-style]

Q2. [NAT · 2 marks]

For the naive recursive Fibonacci function `fib(n)=fib(n-1)+fib(n-2)`, with base cases `fib(0)=0`, `fib(1)=1`, what is the maximum recursion depth when computing `fib(6)`?

Answer: 6

Solution: The deepest chain of nested calls follows the left branch repeatedly: `fib(6)→fib(5)→fib(4)→fib(3)→fib(2)→fib(1)`. That is 6 activation records present on the stack simultaneously at the deepest point.

[GATE-style]

Q3. [MCQ · 1 mark]

The maximum recursion depth reached during execution corresponds to:

- (A) The total number of calls made overall, including ones that have already returned
- (B) The maximum number of activation records simultaneously present on the call stack
- (C) The number of base-case calls only
- (D) The time complexity of the function

Answer: (B)

[GATE-style]

Q4. [NAT · 1 mark]

Consider `f(n){ if(n==0) return 0; return 1+f(n-1); }`. What is the maximum recursion depth (counting `f(10)` itself as depth 1) when calling `f(10)`?

Answer: 11

Solution: Calls made: `f(10),f(9),f(8),...,f(0)` -- that is 11 nested calls in total before the base case returns.

[GATE-style]

Q5. [MCQ · 1 mark]

Which of the following best explains why unbounded (excessively deep) recursion can crash a program?

- (A) The heap runs out of memory
- (B) The call stack (a fixed-size memory region) overflows as activation records accumulate
- (C) The CPU cache becomes corrupted
- (D) Integer overflow always occurs

Answer: (B)

[GATE-style]

Q6. [NAT · 2 marks]

Consider mutual recursion: `isEven(n){ if(n==0) return true; return isOdd(n-1); }`
`isOdd(n){ if(n==0) return false; return isEven(n-1); }`. What is the recursion depth (number of nested calls, counting the initial call) when evaluating `isEven(5)`?

Answer: 6

Solution: `isEven(5)→isOdd(4)→isEven(3)→isOdd(2)→isEven(1)→isOdd(0)` -- the base case is reached inside `isOdd(0)`, giving 6 nested calls total.

[GATE-style]

Q7. [MCQ · 2 marks]

Using `gcd(a,b){ if(b==0) return a; return gcd(b, a%b); }`, how many recursive calls (including the initial call) are made in total while computing `gcd(48,18)`?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

Answer: (B)

Solution: `gcd(48,18) → gcd(18,12) → gcd(12,6) → gcd(6,0)` → returns 6. That is 4 calls in total.

[GATE-style]

Q8. [MSQ · 2 marks]

Which of the following are TRUE about the call stack during recursive execution?

- (A) Each recursive call pushes a new activation record (stack frame) with local variables and return address
- (B) Stack frames are removed (popped) in LIFO order as calls return
- (C) All recursive calls share exactly one single stack frame throughout execution
- (D) A stack frame typically stores the function's parameters for that particular invocation

Answer: (A), (B), (D)

Solution: (C) is false -- each active (not-yet-returned) call has its own distinct stack frame.

[GATE-style]

Q9. [NAT · 2 marks]

Consider `count(n){ if(n==0) return 1; return count(n-1)+count(n-1); }`. What is the maximum recursion depth (the longest single chain from the initial call to a base case) when evaluating `count(7)`?

Answer: 8

Solution: Every branch reduces `n` by exactly 1 down to 0, so the longest chain is `count(7)→count(6)→...→count(0)`, i.e. 8 nested calls -- even though the total number of calls made overall is exponential.

[GATE-style]

Q10. [MCQ · 1 mark]

If a recursive function reaches a maximum recursion depth of d , and each activation record occupies k bytes of stack space, the peak stack memory used is approximately:

- (A) k/d
- (B) $k+d$
- (C) $k \times d$
- (D) $k^2 \times d$

Answer: (C)

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 2 marks]

Consider $f(n)$ { if($n \leq 1$) return 1; return $f(n-1)+f(n-2)$; } (with base cases $f(0)=f(1)=1$). Find the recursion depth (longest chain of nested calls, counting the initial call) when calling $f(8)$.

Answer: 8

Solution: The longest path is $f(8) \rightarrow f(7) \rightarrow f(6) \rightarrow f(5) \rightarrow f(4) \rightarrow f(3) \rightarrow f(2) \rightarrow f(1)$, which is 8 nested calls.

[New Practice]

Q12. [MCQ · 1 mark]

Which of the following recursive patterns has recursion depth $O(\log n)$ rather than $O(n)$?

- (A) $f(n) = f(n-1) + 1$
- (B) $f(n) = f(n/2) + 1$ (binary-search-style halving)
- (C) $f(n) = f(n-1) + f(n-2)$
- (D) $f(n) = n \times f(n-1)$

Answer: (B)

[New Practice]

Q13. [NAT · 1 mark]

For recursive binary search on a sorted array of size 16, what is the maximum recursion depth (number of nested calls, including the initial call, until the base case)?

Answer: 5

Solution: Each call roughly halves the array: sizes $16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$, giving $\log_2(16)=4$ halvings plus the initial call itself, i.e. 5 nested calls.

[New Practice]

Q14. [MCQ · 2 marks]

A recursive function that calls itself twice per invocation (like naive recursive Fibonacci) has, for input n , a maximum recursion depth of:

- (A) $O(2^n)$
- (B) $O(n)$
- (C) $O(\log n)$
- (D) $O(n^2)$

Answer: (B)

Solution: Although the total number of calls made is exponential, the longest single path down one branch still only decreases n by 1 (or more) at each step, so the maximum depth is only $O(n)$.

[New Practice]

Q15. [NAT · 2 marks]

Consider $f(n)$ { if($n \leq 0$) return; print(n); $f(n-3)$; }. What is the recursion depth (number of calls, including the initial one) when calling $f(10)$?

Answer: 5

Solution: $f(10) \rightarrow f(7) \rightarrow f(4) \rightarrow f(1) \rightarrow f(-2)$, where $f(-2)$ is the base case ($n \leq 0$). That is 5 calls in total.

[New Practice]

Q16. [MCQ · 1 mark]

Which statement correctly distinguishes recursion depth from time complexity?

- (A) Recursion depth always equals the time complexity of the function
- (B) Recursion depth measures peak stack usage (space), while time complexity measures total work done; the two can differ substantially (e.g. exponential time but only linear depth)
- (C) Recursion depth is irrelevant to memory usage
- (D) Time complexity is always $\leq$ recursion depth

Answer: (B)

[New Practice]

Q17. [NAT · 2 marks]

Using $\text{gcd}(a,b)$ { if($b==0$) return a ; return $\text{gcd}(b,a\%b)$; }, how many total recursive calls (including the initial call) are made while computing $\text{gcd}(105,45)$?

Answer: 3

Solution: $\text{gcd}(105,45) \rightarrow \text{gcd}(45,15) \rightarrow \text{gcd}(15,0) \rightarrow$ returns 15. That is 3 calls in total.

[New Practice]

Q18. [NAT · 1 mark]

If a recursive function's activation record occupies 100 bytes and the function recurses to a depth of 1000 before hitting its base case, what is the approximate peak stack memory used, in bytes?

Answer: 100000

Solution: Peak memory $\approx$ depth $\times$ frame size = $1000 \times 100 = 100,000$ bytes.

[New Practice]

Q19. [MSQ · 2 marks]

Which of the following can help avoid stack overflow caused by excessive recursion depth?

- (A) Rewriting the recursion as an iterative loop (with an explicit stack, if needed)
- (B) Using tail-call optimization, where the language/compiler supports it
- (C) Always increasing recursion depth without any base case
- (D) Reducing the problem size more aggressively per call (e.g. halving instead of decrementing by 1), where applicable

Answer: (A), (B), (D)

[New Practice]

Q20. [NAT · 2 marks]

Consider $f(n)$ { if($n==0$) return 0; return $1+f(n//2)$; }. What is the recursion depth (number of nested calls, including the initial call) for $f(32)$?

Answer: 7

Solution: $32 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1 \rightarrow 0$ -- calls are $f(32), f(16), f(8), f(4), f(2), f(1), f(0)$, i.e. 7 calls in total.

Topic 6: Recursion Patterns - Factorial, Fibonacci, Recurrence-Writing & GCD

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [NAT · 1 mark]

Using the recurrence $\text{fact}(n) = n \times \text{fact}(n-1)$, $\text{fact}(0)=1$, find $\text{fact}(6)$.

Answer: 720

Solution: $\text{fact}(6) = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$.

[GATE-style]

Q2. [MCQ · 1 mark]

Which recurrence relation correctly defines the factorial function?

- (A) $\text{fact}(n) = \text{fact}(n-1) + n$, $\text{fact}(0)=1$
- (B) $\text{fact}(n) = n \times \text{fact}(n-1)$, $\text{fact}(0)=1$, for $n > 0$
- (C) $\text{fact}(n) = \text{fact}(n-1) \times \text{fact}(n-2)$
- (D) $\text{fact}(n) = \text{fact}(n/2) \times 2$

Answer: (B)

[GATE-style]

Q3. [NAT · 1 mark]

Using $\text{fib}(0)=0$, $\text{fib}(1)=1$, $\text{fib}(n)=\text{fib}(n-1)+\text{fib}(n-2)$, find $\text{fib}(10)$.

Answer: 55

Solution: Sequence: 0,1,1,2,3,5,8,13,21,34,55 for $n=0..10$. So $\text{fib}(10)=55$.

[GATE-style]

Q4. [MCQ · 2 marks]

The time complexity of the naive (non-memoized) recursive Fibonacci implementation $\text{fib}(n)=\text{fib}(n-1)+\text{fib}(n-2)$ is:

- (A) $O(n)$
- (B) $O(n \log n)$
- (C) $O(2^n)$ (exponential)
- (D) $O(n^2)$

Answer: (C)

[GATE-style]

Q5. [NAT · 2 marks]

GCD Problem 1: Using the Euclidean recurrence $\text{gcd}(a,b) = \text{gcd}(b, a \bmod b)$, with $\text{gcd}(a,0)=a$, find $\text{gcd}(56, 98)$.

Answer: 14

Solution: $\text{gcd}(56,98)=\text{gcd}(98,56)$ [since $56 \bmod 98 = 56$] $=\text{gcd}(56,42)$ [$98 \bmod 56=42$] $=\text{gcd}(42,14)$ [$56 \bmod 42=14$] $=\text{gcd}(14,0)$ [$42 \bmod 14=0$] $= 14$.

[GATE-style]

Q6. [NAT · 2 marks]

GCD Problem 2: Find $\gcd(252, 105)$ using the Euclidean recurrence $\gcd(a,b)=\gcd(b, a \bmod b)$, $\gcd(a,0)=a$.

Answer: 21

Solution: $\gcd(252,105) = \gcd(105, 42) [252 \bmod 105 = 42] = \gcd(42, 21) [105 \bmod 42 = 21] = \gcd(21, 0) [42 \bmod 21=0] = 21$.

[GATE-style]

Q7. [NAT · 2 marks]

GCD Problem 3: Find $\gcd(17, 5)$ using the Euclidean recurrence (17 and 5 are co-prime).

Answer: 1

Solution: $\gcd(17,5) = \gcd(5,2) [17 \bmod 5=2] = \gcd(2,1) [5 \bmod 2=1] = \gcd(1,0) [2 \bmod 1=0] = 1$.

[GATE-style]

Q8. [MCQ · 1 mark]

Which recurrence correctly defines the Euclidean algorithm for GCD?

- (A) $\gcd(a,b) = \gcd(a, b-1)$
- (B) $\gcd(a,b) = \gcd(b, a \bmod b)$, with $\gcd(a,0) = a$
- (C) $\gcd(a,b) = \gcd(a-b, b)$ only
- (D) $\gcd(a,b) = a \times b$

Answer: (B)

[GATE-style]

Q9. [MSQ · 2 marks]

Which of the following recurrence relations correctly model the standard recursive definitions stated?

- (A) Factorial call-depth: $T(n) = T(n-1) + O(1)$, $T(0) = O(1)$
- (B) Naive Fibonacci: $T(n) = T(n-1) + T(n-2) + O(1)$
- (C) Euclidean GCD: $T(n) = T(n-1) + O(1)$ (roughly linear number of steps, in the size of the input)
- (D) All recursive functions share the identical recurrence relation $T(n) = T(n-1)$

Answer: (A), (B), (C)

Solution: (D) is false -- different recursive algorithms have very different recurrence structures (linear, exponential, logarithmic, etc.).

[GATE-style]

Q10. [NAT · 1 mark]

Given the recurrence $T(n) = T(n-1) + 1$, $T(0) = 0$, find $T(6)$.

Answer: 6

Solution: This recurrence solves to $T(n)=n$ directly, so $T(6)=6$.

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 1 mark]

Compute $\text{fact}(7)$ using $\text{fact}(n) = n \times \text{fact}(n-1)$, $\text{fact}(0)=1$.

Answer: 5040

Solution: $\text{fact}(7) = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$.

[New Practice]

Q12. [MCQ · 1 mark]

Which base case is sufficient for the standard factorial recurrence $\text{fact}(n) = n \times \text{fact}(n-1)$?

- (A) $\text{fact}(1) = 1$ is the only base case ever needed, regardless of how the recursion is written
- (B) A single base case, $\text{fact}(0)=1$ (or equivalently $\text{fact}(1)=1$), is sufficient
- (C) $\text{fact}(n) = n$ for all n
- (D) No base case is needed since the recursion always terminates on its own

Answer: (B)

[New Practice]

Q13. [NAT · 1 mark]

Using $\text{fib}(0)=0$, $\text{fib}(1)=1$, $\text{fib}(n)=\text{fib}(n-1)+\text{fib}(n-2)$, find $\text{fib}(12)$.

Answer: 144

Solution: Sequence: 0,1,1,2,3,5,8,13,21,34,55,89,144 for $n=0..12$. So $\text{fib}(12)=144$.

[New Practice]

Q14. [MCQ · 2 marks]

Memoization (caching previously computed Fibonacci values) improves the time complexity of recursive Fibonacci from:

- (A) $O(n)$ to $O(1)$
- (B) $O(2^n)$ to $O(n)$
- (C) $O(n^2)$ to $O(n \log n)$
- (D) $O(n)$ to $O(n^2)$

Answer: (B)

[New Practice]

Q15. [NAT · 2 marks]

GCD Problem A: Find $\text{gcd}(1071, 462)$ using the Euclidean algorithm.

Answer: 21

Solution: $\text{gcd}(1071, 462) = \text{gcd}(462, 147)$ [$1071 \bmod 462 = 147$] = $\text{gcd}(147, 21)$ [$462 \bmod 147 = 21$] = $\text{gcd}(21, 0)$ [$147 \bmod 21 = 0$] = 21.

[New Practice]

Q16. [NAT · 1 mark]

GCD Problem B: Find $\text{gcd}(35, 14)$ using the Euclidean algorithm.

Answer: 7

Solution: $\text{gcd}(35, 14) = \text{gcd}(14, 7)$ [$35 \bmod 14 = 7$] = $\text{gcd}(7, 0)$ [$14 \bmod 7 = 0$] = 7.

[New Practice]

Q17. [NAT · 1 mark]

GCD Problem C: Find $\text{gcd}(100, 75)$ using the Euclidean algorithm.

Answer: 25

Solution: $\text{gcd}(100,75) = \text{gcd}(75,25) [100 \bmod 75=25] = \text{gcd}(25,0) [75 \bmod 25=0] = 25$.

[New Practice]

Q18. [MCQ · 2 marks]

Let $M(n)$ denote the number of multiplications performed by the recursive function $\text{fact}(n)$ { if $(n \leq 1)$ return 1; return $n \cdot \text{fact}(n-1)$; } for $n > 0$. Which recurrence correctly describes $M(n)$?

- (A) $M(n) = M(n-1) + 1$, with $M(0) = 0$
- (B) $M(n) = M(n-1) \times n$
- (C) $M(n) = 2M(n-1)$
- (D) $M(n) = M(n-1) + n$

Answer: (A)

Solution: Each call $\text{fact}(n)$ ($n > 0$) performs exactly one multiplication ($n \times \text{fact}(n-1)$), on top of whatever multiplications happened inside the recursive call -- so $M(n) = M(n-1) + 1$, which solves to $M(n) = n$.

[New Practice]

Q19. [NAT · 1 mark]

Using the recurrence $M(n) = M(n-1) + 1$, $M(0) = 0$ from the previous question, find $M(10)$, the number of multiplications performed by $\text{fact}(10)$.

Answer: 10

Solution: $M(n) = n$ directly, so $M(10) = 10$.

[New Practice]

Q20. [MCQ · 2 marks]

Let $B(n)$ denote the number of times a base case ($\text{fib}(0)$ or $\text{fib}(1)$) is directly invoked while naively computing $\text{fib}(n)$ recursively. Which recurrence correctly describes $B(n)$?

- (A) $B(n) = B(n-1) + B(n-2)$, with $B(0) = 1$, $B(1) = 1$
- (B) $B(n) = B(n-1) + 1$
- (C) $B(n) = 2B(n-1)$
- (D) $B(n) = B(n-1) \times B(n-2)$

Answer: (A)

Solution: Every leaf of the Fibonacci recursion tree is a base-case call, and the recursion tree itself branches exactly as $\text{fib}(n)$ does -- so the count of base-case calls follows the identical Fibonacci-style recurrence.

Topic 7: Uninformed Search - BFS: Mechanism & Properties

Section A — Old / Previous-Year-Style Questions (10)

[GATE-style]

Q1. [MCQ · 1 mark]

Breadth-First Search (BFS) explores nodes of a search tree/graph in which order?

- (A) Deepest node first (LIFO order)
- (B) Level by level -- all nodes at depth d are expanded before any node at depth $d+1$ (FIFO order)
- (C) In a random order
- (D) Ordered by a heuristic function value

Answer: (B)

[GATE-style]

Q2. [MCQ · 1 mark]

BFS is implemented using which data structure to manage its frontier (the set of generated-but-not-yet-expanded nodes)?

- (A) A stack (LIFO)
- (B) A queue (FIFO)
- (C) A priority queue ordered by path cost
- (D) A single variable holding one node

Answer: (B)

[GATE-style]

Q3. [MSQ · 2 marks]

Which of the following properties hold for BFS on a search tree with finite branching factor b , when all step (edge) costs are equal?

- (A) BFS is complete (guaranteed to find a solution if one exists)
- (B) BFS is optimal for finding the shortest path when all step costs are equal
- (C) BFS has worst-case time complexity $O(b^d)$, where d is the depth of the shallowest goal
- (D) BFS always uses less memory than DFS

Answer: (A), (B), (C)

Solution: (D) is false -- BFS typically needs far more memory than DFS, since it must store the entire frontier (potentially all nodes at the current depth) simultaneously.

[GATE-style]

Q4. [MCQ · 1 mark]

The space complexity of BFS (number of nodes that must be stored) is:

- (A) $O(bd)$
- (B) $O(b^d)$
- (C) $O(d)$
- (D) $O(1)$

Answer: (B)

Solution: BFS must retain every generated node (frontier + explored set), giving the same exponential order as its time complexity.

[GATE-style]

Q5. [NAT · 2 marks]

A search tree has branching factor $b=2$, and the goal is found at depth $d=4$. In the worst case (full expansion through depth 4), how many total nodes are generated by BFS, i.e. $2^0+2^1+2^2+2^3+2^4$?

Answer: 31

Solution: $1+2+4+8+16 = 31$.

[GATE-style]

Q6. [MCQ · 1 mark]

BFS is guaranteed to find the optimal (shortest-path) solution when:

- (A) All step/edge costs are equal (or the graph is unweighted)
- (B) The heuristic used is admissible
- (C) The graph has negative edge weights
- (D) The search space is a tree only, never a general graph

Answer: (A)

[GATE-style]

Q7. [MCQ · 1 mark]

In BFS on a graph, a node is typically marked "visited"/"explored":

- (A) Only after the goal test fails on it
- (B) As soon as it is generated and added to the frontier, to avoid enqueueing it more than once
- (C) Never -- BFS revisits nodes freely
- (D) Only after all of its neighbours have themselves been fully processed

Answer: (B)

[GATE-style]

Q8. [NAT · 2 marks]

BFS starts at node S, which has 3 neighbours; each of those 3 neighbours has 2 further, non-overlapping, unvisited neighbours of its own. Counting S itself, level 1, and level 2, how many total nodes are visited by the time BFS finishes exploring these two full levels?

Answer: 10

Solution: Level 0: 1 (S). Level 1: 3. Level 2: $3 \times 2 = 6$. Total = $1+3+6 = 10$.

[GATE-style]

Q9. [MCQ · 1 mark]

Compared with Depth-First Search (DFS), BFS is generally preferred when:

- (A) Memory is extremely limited and the solution is expected to be very deep
- (B) We need to guarantee finding the shortest path in an unweighted graph
- (C) The search space is infinite in depth and path length does not matter
- (D) We want the smallest possible memory footprint

Answer: (B)

[GATE-style]

Q10. [MCQ · 1 mark]

If the branching factor $b=1$ (a simple chain-like search space), the time and space complexities of BFS reduce, in terms of depth d , to:

- (A) $O(1)$
- (B) $O(d)$
- (C) $O(d^2)$
- (D) $O(2^d)$

Answer: (B)

Section B — New Practice Questions (10)

[New Practice]

Q11. [NAT · 2 marks]

A tree has branching factor $b=3$, and the shallowest goal is at depth $d=3$. Using BFS's worst-case node generation formula ($b^0 + b^1 + \dots + b^d$), compute the total number of nodes generated.

Answer: 40

Solution: $1+3+9+27 = 40$.

[New Practice]

Q12. [MCQ · 1 mark]

BFS explores the search space using which strategy?

- (A) Depth-first, backtracking whenever stuck
- (B) Breadth-first, expanding all nodes at the current depth before moving deeper
- (C) Best-first, guided by a heuristic function
- (D) Random restarts

Answer: (B)

[New Practice]

Q13. [MSQ · 2 marks]

Which of the following are TRUE regarding BFS's completeness and optimality?

- (A) BFS is complete provided the branching factor is finite
- (B) BFS is optimal only when all path costs are non-negative and equal (or the graph is unweighted)
- (C) BFS is always optimal, regardless of the edge weights
- (D) On an infinite search space with no goal present, BFS may fail to terminate unless depth is bounded

Answer: (A), (B), (D)

Solution: (C) is false -- with unequal edge weights, plain BFS can return a path with fewer edges but higher total cost; Uniform-Cost Search is needed to guarantee optimality in that case.

[New Practice]

Q14. [NAT · 2 marks]

A start node S has branching factor 4, and the goal exists at depth 2. Using BFS's worst-case node generation formula $1+b+b^2$, how many nodes are generated?

Answer: 21

Solution: $1+4+16 = 21$.

[New Practice]

Q15. [MCQ · 1 mark]

Which auxiliary data structure does BFS typically use to avoid processing the same node more than once on a general graph (as opposed to a tree, which has no cycles)?

- (A) A stack of visited nodes
- (B) A visited/explored set (e.g. a hash set or boolean array)
- (C) No such structure is needed for graphs
- (D) A priority queue of visited nodes

Answer: (B)

[New Practice]

Q16. [MCQ · 1 mark]

Expressed in terms of the number of vertices V and edges E (using an adjacency list), the time complexity of BFS graph traversal is:

- (A) $O(V)$
- (B) $O(V+E)$
- (C) $O(V \times E)$
- (D) $O(E^2)$

Answer: (B)

[New Practice]

Q17. [NAT · 1 mark]

Running BFS from a source vertex in an unweighted graph, if the shortest path (in number of edges) from the source to a target vertex is 5, what value does BFS's distance array store for that target vertex?

Answer: 5

Solution: BFS explores nodes in non-decreasing order of distance from the source, so it naturally computes the correct shortest-path distance (in edges) for unweighted graphs.

[New Practice]

Q18. [MCQ · 1 mark]

What is a key difference between running BFS on a tree versus on a general graph?

- (A) There is no difference at all
- (B) On a general graph a visited-set is required to avoid infinite loops caused by cycles, which is unnecessary on a (cycle-free) tree
- (C) BFS cannot be applied to general graphs, only trees
- (D) BFS on a graph always uses a stack instead of a queue

Answer: (B)

[New Practice]

Q19. [MSQ · 2 marks]

Which of the following statements about BFS's frontier (queue) are TRUE?

- (A) The frontier holds all generated-but-not-yet-expanded nodes
- (B) Nodes are removed from the frontier in FIFO order
- (C) The frontier can be implemented with a stack and still behave identically to a queue
- (D) In the worst case, the frontier can hold $O(b^d)$ nodes at the deepest level before it starts shrinking

Answer: (A), (B), (D)

Solution: (C) is false -- replacing the queue (FIFO) with a stack (LIFO) changes the algorithm into DFS, destroying BFS's level-by-level guarantee.

[New Practice]

Q20. [NAT · 1 mark]

Using BFS on a binary tree (branching factor $b=2$) to find a goal at depth $d=5$, how many nodes exist at depth 5 alone (i.e. b^d)?

Answer: 32

Solution: $2^5 = 32$.
